Portable Causal Fairness: Graph Surgery Across Generator Families

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Abstract

Institutions that release synthetic data in place of sensitive records, such as a hospital, a lending regulator or a statistical agency, choose the process that generates the table, and can shape that process so unfair pathways are absent. DECAF made this concrete for one GAN: factor the generator along a causal graph, sever the edges from a protected attribute to an outcome, and three fairness definitions become three sets of edge cuts. But DECAF is a single GAN, carrying no privacy guarantee and no competitive data-quality story, and the mechanism has never been shown to work in any other architecture. We ask whether it belongs to DECAF or to causal factorisation itself, porting all three definitions to nine generators from three unrelated families (marginals-based, GAN-based and diffusion-based, each with differentially private variants) across 2,520 matched-pair runs on the Adult and COMPAS datasets. The definitions transfer everywhere: cutting edges reduces downstream disparity in every family, dataset and privacy level. The utility cost is small (under two accuracy points for the cuts worth making in five of six family–dataset cells), and the privacy cost is nil in the marginals family, because a severed edge is never measured under the privacy budget. In this setting privacy and fairness turn out to behave close to independently rather than adversarially. Causal fairness is now available inside a modern differentially private synthesizer and inside a diffusion backbone, not only inside one GAN.

1 Introduction

Institutions holding sensitive records increasingly publish a synthetic substitute rather than the records themselves: a hospital, a lending regulator or a statistical agency fits a generative model and releases a table drawn from it. That substitution is usually justified on privacy grounds, but it carries a second, less-exploited consequence. Whoever generates the data chooses the generating process, and a process can be edited. A disparity that exists in the world because of a pathway running from a protected attribute, such as national origin, to an outcome, such as whether a loan is approved, need not exist in the released table.

The cleanest existing realisation of that lever is DECAF [van Breugel et al., 2021], which factors a generator along a causal graph so that severing an edge at sampling time makes the generator structurally unable to use the pathway it named. But DECAF is a single non-private WGAN-GP [Gulrajani et al., 2017], whose released tables trail modern differentially private marginals-based synthesizers and modern diffusion models on the fidelity and utility benchmarks that decide what institutions actually deploy. Whether the fairness mechanism is confined to that one architecture, or is a property of causal factorisation itself, is therefore a practical question and not an academic one: it decides whether an institution deploying synthetic data has the intervention available at all.

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We ask two questions of that mechanism. Is edge surgery a property of DECAF, or of the causal factorisation any such generator uses? And can it be attached to a generator whose graph is not the directed causal DAG DECAF was written against, but the undirected marginal-dependency graph a differentially private synthesizer builds from noisy counts? The stakes are concrete. A hospital publishing patient outcomes, a regulator publishing loan approvals, a court publishing risk scores each hand analysts a table that will be treated as ground truth by whoever receives it. If the mechanism that would make that table fair is confined to a generator no such institution would deploy, it might as well not exist.

2 Background and related work

DECAF: fairness as graph surgery. DECAF [van Breugel et al., 2021] generates fair synthetic tables from a GAN whose generator is *factored* along a causal DAG (directed acyclic graph): the joint distribution over columns is written as a product of conditionals, $P(X) = \prod_i P(X_i \mid \mathrm{pa}(X_i))$, with $\mathrm{pa}(X_i)$ the columns the graph draws as parents of column X_i. This is similar to the decomposition a marginals-based synthesizer such as MST fits, differing chiefly in that MST's dependency graph is undirected. DECAF gives each column its own sub-network reading only $\mathrm{pa}(X_i)$, so a parent can be severed at generation time by substituting an independently drawn surrogate, leaving the generator structurally unable to use the unfair pathway and requiring no retraining.

Which edges to sever turns a fairness *definition* into an *algorithm*, and DECAF supplies three (Figure 1). **FTU** (fairness through unawareness) cuts only the direct edge $S \to Y$; **DemPar**[1] (demographic parity) cuts every directed path $S \rightsquigarrow Y$; **CF** (conditional fairness) spares paths that pass through an attribute the modeller has nominated *admissible*, so disparity routed through a legitimate factor survives. DECAF's own illustration is the clearest: *Education* $\to$ *Resume* $\to$ *Job* is worth keeping because *Resume* is admissible, while *Race* $\to$ *Postcode* $\to$ *Loan* is redlining and is not. The graph-cutting machinery is unusually clean, and its authors argue it should port to other generators.

Differential privacy. Institutions releasing sensitive tables increasingly attach a differential privacy (DP) guarantee [Dwork et al., 2006], which caps the influence any single individual can have on the released synthetic table by a budget ε: smaller ε means stronger privacy and noisier releases. ε=1 is the tight budget an institution might publish under; ε=1000 is a permissive comparison point that approximates no privacy at all. Every mention of "private" or "DP-" below refers to such a synthesizer.

Related approaches. Only one prior system attaches an in-generator fairness cut to a private synthesizer. PreFair [Pujol et al., 2023] adapts the CF definition above to MST [McKenna et al., 2021], excluding offending marginals from candidacy before the private structure search runs; it implements only that one definition, and is neither built nor evaluated for any synthesizer other than MST. Causal-DAG diffusion generates column by column much as DECAF does but targets fidelity, with no protected attribute and no edge removal [Zhang et al., 2025, Jacob et al., 2026]. Private tabular diffusion without a causal graph exposes no per-column edge to cut and cannot host the mechanism at all [Zhu et al., 2024]. Everything else that combines privacy with fairness for generation acts outside the graph: enforcing statistical independence on a learned intermediate embedding [Hyrup et al., 2025, Sarmin et al., 2025], prompting an LLM backbone [Nagesh et al., 2025], learning a private structure without a fairness objective [Jia et al., 2025], or mitigating downstream of the synthesizer [Angelozzi and Arcolezi, 2026].

The gap. Nobody has asked whether DECAF's edge surgery is a property of *DECAF* or of *causal factorisation*. DECAF conjectures the latter twice, calling the method "simple and extendable to other generative methods" and "directly extendable to the more stable WGAN-GP and other generative models", but tests neither claim. If the conjecture holds, the mechanism should transfer to any generator factored along a graph, including the marginals-based synthesizers that dominate private tabular data [McKenna et al., 2021, Zhang et al., 2014, 2021, McKenna et al., 2022] and including diffusion, the current tabular quality leader [Kotelnikov et al., 2023].

[1] DECAF calls this mechanism DP; we write DemPar throughout and reserve DP for differential privacy, discussed alongside it in every result below.

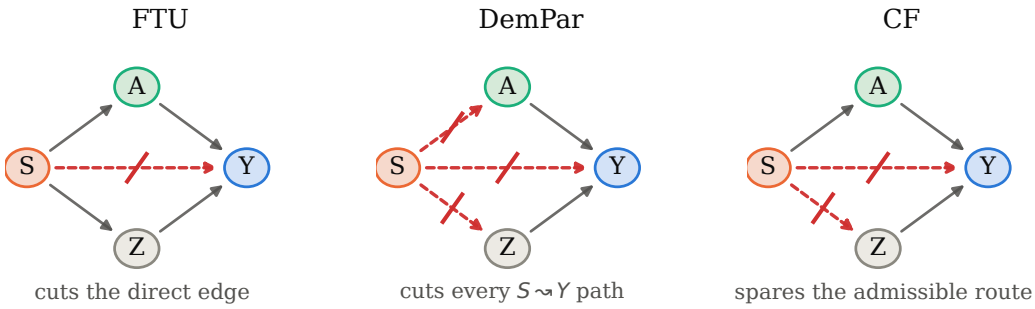


Figure 1: Three fairness definitions as three sets of severed edges, acting on the graph the data is *generated* from, not on the model trained afterwards. S is the protected attribute, Y the outcome, A an admissible attribute whose influence is deemed legitimate, and Z an intermediate proxy. Dashed red edges are cut. DemPar removes every directed $S \rightsquigarrow Y$ path; CF removes only those that avoid A, so the route $S \to A \to Y$ survives.

3 Porting the mechanisms

We implement FTU, DemPar and CF against one interface and attach it to nine generators from three unrelated families. The definitions do not change; only the graph the cut applies to does, and whether that graph is directed or undirected.

Marginals-based synthesizers (MST, PrivBayes, PrivSyn, and AIM[2]) form a graphical model whose edges are the pairs of columns whose noisy joint counts the synthesizer chooses to measure under the privacy budget; the release is drawn from a distribution that reproduces those counts. Because the graph is undirected, the fairness cut is enforced on paths rather than on ordered parent→child edges: whether adding a candidate pair would create a forbidden S-to-Y path depends on what is already in the model, so the constraint is checked incrementally at each selection step. A pair the cut excludes is never measured, making the constraint free in the privacy accounting rather than merely cheap.[3] **Causal GANs** (DECAF, plus CTGAN-based [Xu et al., 2019] and DP-SGD [Abadi et al., 2016] variants) use DECAF's surrogate substitution unchanged, on the directed DAG. **Causal diffusion** keeps the same factorisation and replaces each sub-network with a small denoiser trained by supervised noise prediction; the cut is the same code path, because what exposes an edge is that X_i has its own network reading only $\mathrm{pa}(X_i)$, true whether that network is trained adversarially or by denoising.

Experimental setup. Two datasets, Adult [Becker and Kohavi, 1996] and COMPAS [Angwin et al., 2016]; nine generators; four mechanisms (NONE/FTU/DemPar/CF); $\varepsilon \in \{1, 10, 1000\}$ at $\delta = 10^{-9}$ for the six generators carrying a differential-privacy guarantee [Dwork et al., 2006]; three protected/admissible role splits per dataset; five seeds; 2,520 completed runs. One Adult split reproduces DECAF's own setup (protected `sex`; admissible `occupation`, `hours-per-week`, `workclass`, `education`), one follows PreFair's Table 1, and one is deliberately strict; COMPAS is absent from DECAF's evaluation, so its splits follow PreFair and two variants. For the causal GAN and diffusion families the DAG for each dataset is specified by hand from domain knowledge, as is standard in this literature (following Kusner et al. [2017] and Zhang et al. [2017]) and held fixed across generators so effects are attributable to the mechanism. The marginals-based generators choose their own undirected graph under the privacy budget, and the cut is enforced on that graph.

Metrics. *Fidelity* is 1-way and 2-way total-variation distance between the synthetic and real marginals, averaged over columns and column-pairs (lower is better; a value of 0 means the synthetic and real marginals agree exactly). *Utility* is the test accuracy of an MLP trained on the syn-

[2]AIM [McKenna et al., 2022] was ported after the main sweep and run at ε=1 with one seed per (dataset, mechanism); it is included as a smoke-test on a fourth marginals backbone, not pooled into the Table 1 averages.

[3]The freed ε does not buy the release a better fit. MST fixes its per-round budget from the number of edges a spanning tree needs, before selection begins, so pruning candidates shrinks each round's pool rather than enriching what remains, and matched-pair 2-way TVD is worse under the cut, by 0.008 for DemPar, 0.003 for CF and 0.001 for FTU on Adult (ordered by how much structure each removes).

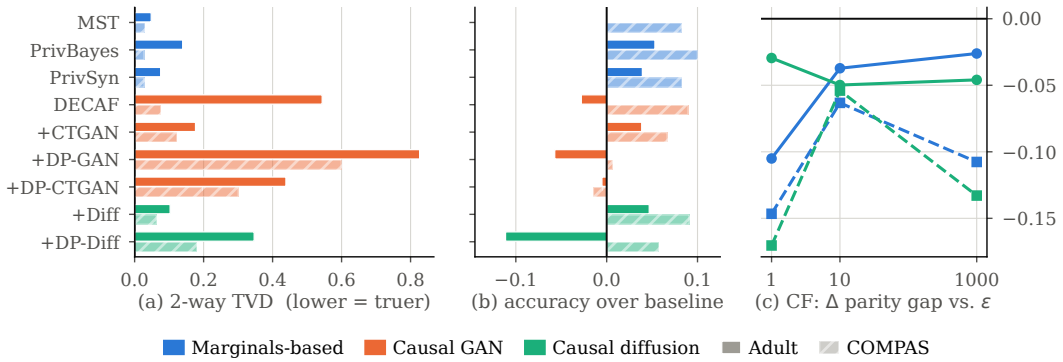


Figure 2: **(a, b)** every generator with no fairness mechanism, averaged over ε, role splits and seeds; (a) 2-way TVD to the real table (lower is better), (b) MLP lift over the majority-class rate (higher is better). A bar at zero in (b) means the data taught a classifier nothing, and a table with no signal has no disparity left to remove. **(c)** the matched-pairs reduction in parity gap from CF at each budget, for the two families present in every cell (marginals and diffusion; the causal GAN family lacks a matched cell at every budget because DP-GAN and DP-CTGAN collapse to a constant predictor in several cells). More negative is fairer, and the effect does not decay as ε tightens.

thetic table to predict the outcome column (`income` on Adult, `two_year_recid` on COMPAS) and evaluated on a real held-out 30%; we report it as *lift* over the majority-class rate so that a generator with no learnable signal scores zero rather than the majority baseline. *Fairness* is that same MLP's worst-case demographic-parity gap across values of the protected attribute, again evaluated against real held-out people (DECAF's DF-ORIGINAL reference). Concretely, if 50% of men and 17% of women in the released table are predicted to earn above \$50k, the demographic-parity gap on Adult is $|0.50 - 0.17| = 0.33$; a mechanism that reduces it to 0.05 has removed 85% of the gap. We also compute conditional parity and positive/negative-rate balance; every claim below reproduces on all four. Seed noise rivals the effects themselves (median within-cell SD of the parity gap: 0.031 on Adult, 0.092 on COMPAS), so every fairness number below is a **matched-pairs** difference: each run with a fairness mechanism is differenced against the run identical in dataset, generator, ε, role split *and seed* but with no fairness mechanism.

4 Results

4.1 Fidelity and utility

Fidelity separates the nine generators cleanly (Figure 2a). The marginals-based synthesizers dominate, as expected since they optimise almost exactly 2-way TVD: MST reaches 0.048 on Adult and 0.030 on COMPAS. Diffusion is the next tier and holds it at every privacy level: our non-private diffusion port reaches 0.103 on Adult against DECAF's own 0.544, and with DP-SGD attached, DP-diffusion at 0.346 still leads DP-GAN at 0.439 and DP-CTGAN at 0.830.

Utility is the load-bearing panel for what follows (Figure 2b). Most generators clear the majority-class baseline meaningfully: on Adult PrivBayes reaches 0.05 lift over the 0.758 majority-class rate, PrivSyn 0.05, our non-private diffusion port 0.04, and DP-diffusion at $\varepsilon \geq 10$ sits just above baseline. Two cells are trouble: MST on Adult sits at exactly zero lift (accuracy 0.747, majority rate 0.758), and the DP-GAN family collapses to a constant predictor in several ε=1 cells. A table that teaches a classifier nothing has no disparity for a fairness mechanism to remove, and is scored as perfectly fair by every existing fairness metric. Every fairness claim below is a matched-pair difference between mechanisms holding the generator fixed, so this artefact does not confound the mechanism's effect, but it does mean an absolute fairness number reported without the utility number beside it is uninterpretable for those two cells specifically.

Table 1: Matched-pairs change in the worst-case demographic-parity gap from applying each mechanism, pooled within generator family. *end* is the gap that remains after the mechanism; Δ gap is the mean paired change with a 95% CI (negative = gap shrunk); Δ acc is the paired MLP-accuracy change. Base (no-mechanism) gaps are in the family headers.

	Adult			**COMPAS**		
Mech.	end	Δ gap	Δ acc	end	Δ gap	Δ acc
Marginals-based (MST, PrivBayes, PrivSyn), base gap 0.143 / 0.331, $n = 135$						
FTU	0.121	-0.022 ± 0.020	$+0.002$	0.254	-0.076 ± 0.027	$+0.003$
CF	0.087	-0.056 ± 0.020	-0.005	0.225	-0.106 ± 0.027	-0.011
DemPar	0.071	-0.073 ± 0.021	-0.032	0.230	-0.101 ± 0.029	-0.033
Causal GAN (DECAF, +CTGAN, +DP-GAN, +DP-CTGAN), base gap 0.074 / 0.238, $n = 102/96$						
FTU	0.072	-0.003 ± 0.006	-0.000	0.156	-0.081 ± 0.028	-0.002
CF	0.019	-0.055 ± 0.019	-0.001	0.134	-0.103 ± 0.031	-0.017
DemPar	0.007	-0.068 ± 0.021	-0.003	0.094	-0.145 ± 0.039	-0.049
Causal diffusion (DECAF+Diffusion, +DP-Diffusion), base gap 0.069 / 0.294, $n = 60$						
FTU	0.059	-0.010 ± 0.014	-0.004	0.193	-0.101 ± 0.029	-0.013
CF	0.015	-0.055 ± 0.017	-0.015	0.160	-0.134 ± 0.034	-0.033
DemPar	0.003	-0.066 ± 0.016	-0.007	0.126	-0.167 ± 0.034	-0.088

4.2 Fairness: the definitions transfer, the magnitude does not

The definitions transfer. In every generator family, on both datasets and at every privacy level, cutting edges reduces the downstream parity gap (Table 1: the mean matched-pair reductions are every one signed the same way and every one comfortably outside its own 95% interval). This answers DECAF's own conjecture in the sense that matters first: nothing about the three definitions requires a GAN, and nothing about them requires a supplied directed DAG.

The magnitude does not transfer, and the pooled averages hide that. The pooled numbers look invariant at first: on Adult, CF's mean matched-pair reduction in parity gap is 0.056 ± 0.020 on marginals, 0.055 ± 0.019 on causal GANs and 0.055 ± 0.017 on causal diffusion, three means agreeing to 0.002. The per-generator effects on Adult span 0.002 (MST) to 0.173 (DECAF), and each family happens to contain both a generator that reproduces the disparity in full and a generator whose learned signal has collapsed to the majority class (utility bar at 0 in Figure 2b), so the family means average across those extremes and land on the same intermediate value. The comparison as a fraction of each generator's own baseline parity gap is more informative. Restricted to matched pairs where there is disparity to remove (baseline gap > 0.05) and where the generator learned something (lift > 0.02), CF removes 89% of the baseline parity gap on the causal diffusion family, 75% on the causal GAN family and 29% on the marginals-based family on Adult (43%, 32% and 22% on COMPAS). Concretely, on DECAF diffusion under DemPar on Adult, the fraction of men in the released table that the downstream classifier predicts to earn above \$50k differs from the corresponding fraction for women by 0.29 before the mechanism is applied and 0.03 after. On MST the same mechanism moves it from 0.11 to 0.09. Variance partitioning confirms the split: which family the generator belongs to explains essentially none of the raw reduction ($\eta^2 = 0.00$) and about a quarter of the fraction removed ($\eta^2 = 0.25$; $n = 432$ matched pairs in the disparity-and-signal subset).

The gap has a structural cause: the same definition compiles to two different operations, on very different graphs. The causal GAN and diffusion families receive a directed DAG from a domain expert and sever its edges by surrogate substitution at sampling time, so the cut is post-processing on a supplied graph. A marginals-based synthesizer receives no DAG; it selects an undirected dependency structure under the privacy budget, so the same definition has to be enforced upstream, during that selection, and it can only sever an edge the structure carries in the first place. The two graphs carry very different amounts of protected-to-outcome dependence to cut (Table 2): the hand-designed Adult DAG puts ≈ 9 edges on some S-to-Y path that DemPar must sever, whereas the marginals-family graph, private and sparse by construction, offers only 4. The GAN and diffusion families therefore cut absolutely more edges but relatively less of their (already small) baseline disparity is left to remove; the marginals family cuts fewer edges but a much larger fraction of its baseline gap. The definitions are portable because they are stated over graphs; the magnitude is not, because the two families are not operating on the same graph.

Table 2: Mean number of edges the mechanism must sever, per run, per generator family. For the causal GAN/diffusion families the count is deterministic: it is the number of DAG edges DECAF's surrogate substitution would replace given each dataset's role split. For marginals it is the minimum edge cut needed on the baseline (no-mechanism) learned undirected graph to break every forbidden S-to-Y path (paths through admissibles are legal under CF, hence its intermediate size). DemPar $\geq$ CF $\geq$ FTU by construction.

	Adult			**COMPAS**		
Family	FTU	CF	DemPar	FTU	CF	DemPar
Causal GAN / diffusion (fixed DAG)	0.75	4.00	8.75	1.33	1.67	3.33
Marginals-based (learned graph)	0.58	2.81	4.24	0.96	2.27	3.72

Table 3: **Where to intervene, on identical data.** MST synthetic training data, real held-out test set, XGBoost, 5 seeds, all inside the benchmark code of Angelozzi and Arcolezi [2026]. Entries are |parity gap| / accuracy; **bold** marks the best value in each column at the displayed precision (lower gap, higher accuracy; ties bolded together). The eight comparison mitigators act on the released table or on the model trained from it; our three (**in-gen (ours)**) act inside the generator. Majority-class accuracy is 0.758 (Adult) and 0.533 (COMPAS), so learned fair representations buy their low gap by falling to it.

		Adult		COMPAS	
stage	intervention	ε=1	ε=10	ε=1	ε=10
	no mitigation	0.068 / 0.74	0.073 / 0.74	0.109 / **0.61**	0.190 / 0.61
pre	Reweighing	0.087 / **0.75**	0.085 / **0.75**	0.132 / **0.61**	0.093 / 0.61
pre	Disp. impact rem.	0.078 / 0.74	0.105 / 0.72	0.119 / **0.61**	0.205 / **0.62**
pre	Learned fair repr.	0.018 / 0.72	0.031 / 0.70	0.037 / 0.50	0.028 / 0.51
in	Exp. gradient	0.047 / 0.74	0.058 / 0.74	0.115 / **0.61**	0.053 / 0.60
in	Grid search	0.068 / 0.74	0.073 / 0.74	0.109 / **0.61**	0.190 / 0.61
post	Reject option	0.045 / 0.60	0.046 / 0.61	0.045 / 0.56	0.046 / 0.54
post	Equalised odds	**0.009** / 0.71	0.015 / 0.72	**0.024** / 0.58	**0.023** / 0.59
post	Calib. eq. odds	0.049 / **0.75**	0.040 / **0.75**	0.485 / 0.56	0.415 / 0.57
in-gen (ours)	FTU	0.073 / 0.74	0.062 / 0.74	0.087 / 0.53	0.155 / 0.61
in-gen (ours)	DemPar	0.010 / **0.75**	**0.006** / **0.75**	0.067 / 0.53	0.148 / **0.62**
in-gen (ours)	CF	0.015 / **0.75**	0.029 / 0.74	0.111 / 0.59	0.155 / 0.61

The predicted ordering falls out of the data. Ranked by parity gap left behind, NONE $>$ FTU $>$ CF $>$ DemPar holds in five of six (dataset $\times$ family) cells (Table 1). CF landing between the other two is a substantive check on the mechanism, not a formality: because CF spares disparity routed through the admissible attribute, wherever the role split gives DemPar a mediator to cut and CF one to spare, CF must recover strictly less parity-gap reduction, and wherever the two separate by more than seed noise, it does.

Fairness through unawareness is weak, and on Adult barely acts. Pooled over Adult, FTU shifts the matched-pair parity gap by only 0.013 on average. The pooled figure is the sum of two very different behaviours: on MST the mechanism is a literal no-op in all 45 Adult pairs, because the direct protected-to-income marginal is one the private structure search never buys under our budgets and there is nothing for the filter to exclude; on the GAN and diffusion families FTU does cut, but the influence promptly re-routes through proxy attributes. This is the textbook criticism of unawareness, confirmed constructively: the cheap cut fails where the two path-based cuts, on the same generators and seeds, work.

The utility cost tracks the aggressiveness of the cut, and is modest for the cuts worth making: pooled over Adult, DemPar costs 1.7 accuracy points and CF costs 0.6; on COMPAS, 5.0 and 1.7. CF stays under two accuracy points in five of six (dataset $\times$ family) cells.

4.3 How this compares to intervening downstream

Everything so far measures the mechanism against itself. To place it against the alternatives an institution actually has, we ran the benchmark of Angelozzi and Arcolezi [2026] (introduced in §2),

which asks where in a differentially private synthetic pipeline a fairness intervention should go and offers eight standard answers spread across pre-processing the table, constraining the classifier, and post-processing its predictions. We add ours as a fourth stage, applied inside the generator. We use their code unmodified: their preprocessed Adult and COMPAS, their train/test splits, their MST synthesizer, their XGBoost classifier and their aif360 fairness metrics. The only change is that for our arm the training table was generated by our MST under FTU, DemPar or CF rather than by theirs. Table 3 is the result.

On Adult the in-generator cut is on the fairness/accuracy frontier (Table 3, columns 1–2). DemPar in-generator leaves a parity gap of 0.010 at 0.748 accuracy; the only downstream mitigator that reaches a comparable gap is equalised-odds post-processing (which threshold- shifts the classifier per group) at 0.009, and it costs 3.4 more accuracy points to get there. Reweighing and the disparate-impact remover leave the release *less* fair than no mitigation at all in this cell, and grid search is an exact no-op. On COMPAS the ordering reverses: post-processing reaches 0.023 where our best cut reaches 0.148. This is the same finding as §4 seen from outside, since at these budgets the privately selected MST structure barely encodes the protected-to-outcome dependence and there is correspondingly little for a cut to sever, whereas a mitigator acting on the trained model is not subject to that limit. FTU and CF are exact no-ops on Adult in this harness too, reproducing the FTU result above in a different codebase and metric.

The comparison is not a like-for-like ranking of techniques, because the interventions are not substitutes. A downstream mitigator is applied by whoever trains a model, and a public table has consumers its publisher will never meet; an in-generator cut is applied once by the publisher and is the only one of the four that ships with the released table.

4.4 Privacy and fairness are nearly independent

The standard expectation for privacy and fairness is a trade-off: differential privacy has been shown to fall hardest on the smallest groups [Bagdasaryan et al., 2019], so tightening ε should both widen the parity gap in the released table and blunt any attempt to close it. In this setting we do not see the second half. Cutting edges reduces the parity gap at least as much at the strong $\varepsilon{=}1$ as at the permissive $\varepsilon{=}1000$ in nine of twelve (family $\times$ dataset $\times$ {CF, DemPar}) cells; two of the three exceptions are COMPAS GAN cells where the backbone has already collapsed to a constant predictor at $\varepsilon{=}1$ and has no disparity left to remove. For marginals-based generators on Adult, CF's mean paired reduction in parity gap is 0.105 at $\varepsilon{=}1$ against 0.026 at $\varepsilon{=}1000$ (on COMPAS, 0.147 against 0.108). Part of the reason the reduction is *larger* at $\varepsilon{=}1$ is that DP noise has already widened the baseline parity gap there, so more remains for a cut to remove. That does not explain it all though: the reduction moves the same way as a fraction of baseline (68% against 18% on Adult). For causal diffusion the reduction is essentially flat: on Adult CF removes 0.030, 0.050 and 0.046 across the three budgets. One deviation we cannot yet explain is visible in Figure 2c: the effect at $\varepsilon{=}10$ is weaker than at either neighbour in both families and both datasets.

One private generator fails outright, on a claim DECAF makes. DECAF argues from the post-processing theorem that it "can be directly used for generating private synthetic data by replacing the standard discriminator by a differentially private discriminator." The accounting is correct; the output is not. Attaching DP-SGD [Abadi et al., 2016] to DECAF's WGAN-GP critic in this way (the DP-GAN row in Table 1) collapses the synthetic outcome to a single class in 168 of 360 runs (47%), every COMPAS run at $\varepsilon{=}1$ included. Two independent changes each remove the failure: substituting a CTGAN representation under the same adversarial training ($1/360$), and substituting diffusion for the WGAN-GP critic under DECAF's own representation ($0/360$). Two unrelated repairs locate the cause in DP-SGD on a weight-clipped critic, not in private causal generation as such.

5 Discussion

The cut is only as strong as the graph it is given. DECAF severs edges in a causal DAG that a domain expert supplied, at sampling time. A differentially private marginals-based synthesizer is supplied with nothing: it buys a structure out of the privacy budget, so the same definition has to be enforced during structure selection, against whatever graph the noise could afford. That is why the definitions port cleanly across the three families while their effect sizes do not, and the limiting

case is stark. Fairness through unawareness on MST is a literal no-op in all 45 matched Adult pairs, because the direct `sex`→`income` marginal is one the private structure search never buys under our budgets; the constraint is satisfied before it is imposed. The practical consequence for an auditor is that knowing which generator produced a release tells them little about how much its fairness mechanism did: the graph it sampled along, and which attributes were nominated admissible for CF, are the load-bearing inputs.

The in-generator cut and downstream mitigation address different disparities, and the split predicts which one to use. A downstream mitigator sees the released table and adjusts either it or a model trained on it; the in-generator cut acts on the generative structure so the released table is already what a fair one would look like. Table 3 lets us read that split directly. On Adult at $\varepsilon{=}1$ the disparity is structural — carried by protected-to-outcome edges the generator would otherwise reproduce — and DemPar in-generator arrives at 0.010 gap / 0.748 accuracy, matching the best post-processing on gap and beating it by three accuracy points (0.009 / 0.714); post-processing has nothing left to do. On COMPAS the privately selected structure barely encodes the protected-to-outcome dependence, the in-generator cut has little to sever, and the residual disparity that survives lives in noise the downstream mitigators are built to handle. Publishers who release a public table should prefer the in-generator cut where it applies, because it is the one intervention that ships with the table and cannot be forgotten by a downstream consumer; the two mechanisms are complements when either is missing.

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